

PAPER_1410_POP_III_IMF_TOP_HEAVY

Daniel T. Murphy

PAPER_1410 — UQFF Derivation of Population III Initial Mass Function (CLOSED — EXACT Identity)

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: G — Astrophysics (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: $\text{calculate_paradox}(\{'paradox': 'pop3_imf'\}) \rightarrow A_5 \times (D+1)/(D-1)$
 $= 100 \text{ M_sun EXACT}$

Observation

First-generation (Pop III) stars are predicted to be top-heavy with characteristic mass $\sim 100 \text{ M_sun}$. UQFF derives this exact value.

UQFF Closed Identity

``

$M_{\text{PopIII}} = A_5 \times (D_{\text{phys}} + 1) / (D_{\text{phys}} - 1) = 60 \times 5/3 = 100 \text{ M_sun EXACT}$

``

Physical Interpretation

Icosahedral group order $|A_5| = 60$ scales the structural mass scale; the $(D+1)/(D-1)$ factor encodes the spacetime-volume ratio of the early universe.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF EXACT integer-primitive identity for Population III Initial Mass Function.